

SOC Incident Report

Incident Type: Suspected SMB Brute-Force Authentication Attempt

Environment: Internal SOC Lab (Isolated Virtual Network)

Date: 05 Feb 2026

Analyst: Rishabh Arora

Summary

An SMB authentication attack was simulated from a Kali Linux system against a Windows endpoint within an isolated lab environment. Multiple failed network logon attempts were observed, indicating a potential brute-force or password-spray attempt. The activity was detected using native Windows Security logs and endpoint telemetry.

No successful compromise was identified, and the incident was contained through existing access controls and authentication mechanisms.

Environment Overview

Attacker Machine: Kali Linux (VM)

Victim Machine: Windows 10 (VM)

Network: VMware NAT (isolated lab)

Services Targeted: SMB (TCP 445)

Monitoring Tools:

- Windows Security Logs
- Sysmon (Endpoint Telemetry)

Centralized Logging Architecture

Windows endpoint logs were forwarded to a centralized Ubuntu-based log collection server using NXLog over UDP (port 514).

This enabled centralized visibility, correlation, and investigation of authentication and network-related events across the lab environment.

Attack Description

The attacker system initiated SMB authentication attempts against the Windows host. The attempts targeted standard user and administrator-level accounts using incorrect credentials.

The attack was limited to authentication attempts only and did not involve exploit execution, malware delivery, or lateral movement beyond SMB access attempts.

Detection Method

The activity was detected through analysis of Windows Security logs on the victim machine.

Key Indicators Observed:

- Multiple failed network logon attempts
- Repeated authentication failures from a single source IP
- Network-based logon type consistent with SMB access

Endpoint telemetry confirmed inbound network connections corresponding to the attack timeframe.

Detection Logic

Detection Name: SMB Brute-Force Authentication Attempt

Log Sources:

- Windows Security Logs
- Sysmon Network Telemetry
- Centralized via NXLog to Ubuntu SIEM node

Logic:

IF

Failed logon events are observed

AND

Logon type is Network (SMB-related)

AND

Same source IP generates multiple failures

WITHIN

A short time window

THEN

Flag activity as suspected SMB brute-force attempt

Evidence Collected

Windows Security Logs:

- **Event Type:** Failed authentication events
- **Logon Type:** Network-based logon
- **Source:** External system within internal network
- **Result:** Authentication failure

Endpoint Telemetry (Sysmon):

- Network connections originating from the attacker system
- Correlation between network activity and authentication failures

Impact Assessment

- **Account Compromise:** None
- **Privilege Escalation:** None
- **Data Access:** None
- **Service Disruption:** None

The attack was unsuccessful and did not result in unauthorized access.

Root Cause Analysis

The incident was caused by repeated authentication attempts using invalid credentials against SMB services. Existing authentication controls and access restrictions prevented successful logon.

The activity is consistent with early-stage brute-force or password-spray behavior.

Mitigation & Response

- Authentication failures were logged and reviewed
- No immediate containment actions required due to unsuccessful access
- Continued monitoring recommended for repeated authentication attempts

Recommendations

- Implement alerting for repeated failed network logons from a single source
- Apply account lockout policies to reduce brute-force risk
- Continue endpoint telemetry monitoring for authentication-based attacks

Limitations

This simulation was conducted in an isolated lab environment without domain services.

Detection logic was validated conceptually through log correlation rather than automated alerting in a production SIEM platform.

Conclusion

This incident demonstrated the effectiveness of Windows authentication logging and endpoint telemetry in detecting unauthorized SMB authentication attempts. While no compromise occurred, the activity highlights the importance of monitoring failed logons and correlating network activity for early attack detection.